DSM2274N/2294N System Administration & Management

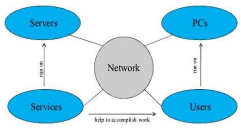
Lec01 - Introduction

Topics

1. What is System Administration?
2. What do system admins do?
3. Principles and First Steps
4. Organizations and Certifications
5. Maturity and Complexity
6. Ethics

What is a System?

- System: An organized collection of computers interacting with a group of users.



What is System Administration

- System administration is the field of work which involves...
  - Managing one or more systems, in their various states
  - System could be software, hardware, Servers or workstations.
  - The objective are efficiency and effectiveness
- System administration matters because computers and networks matter.

  Business Operations | Colloborations | Solve Many Problems
- Business Depends on

  Internet | Intranet | World Wide Web
- Management now has a more realistic view of computers, More people becoming direct users of Computers
- Unreliable machine room power system that caused occasional but bearable problems now prevents sales from being recorded.
- Computers matter more than ever. If computers are to work and work well, system administration matters

Orgnization Size and System Administrtaion

- Small
  - **2-10** computer's 💻, **1** OS, **2-20** user's 👥
  - The system administrator (sys-admin) can be entire IT staff
  - Small staff size may require outsourcing to obtain most specialized skills.
- Mid-sized
  - **11-100** computer's 💻, **1-3** OSes, **21-100** user's 👥
  - These organizations are larger than small businesses but smaller than large enterprises.
- Large
  - **> 100+** computer's 💻, **multiple** OSes, > **100+** user's 👥
  - The sys-admin is one of many IT staff, a specialist instead of "jack of all trades"
  - Other IT roles may include database admin, network admin, fileserver admin, help desk worker, programmers, logistics
  - Outsourcing is done to reduce cost, and for some specializations.

What is a System Administrator?

1. Difficult to define because system administrators do so many things:
   - looks after computers, networks, and the people who use them.
   - look after hardware, operating systems, software, configurations, applications, or security
2. A system administrator influences how effectively other people can or do use their computers and networks.

   Help Employees | Support Others | Guide Others
3. A system administrator sometimes needs to be a

   business-process consultant | corporate visionary | janitor | software engineer | electrical engineer | psychiatrist | mindreader | economist
4. A role that is known by different names

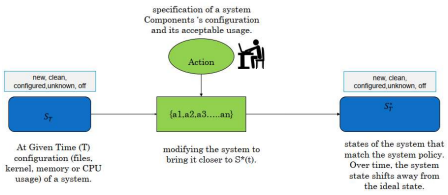
   network administrators | system architects | system engineers | operators | system programmers

What does Managing an IT System means?

- Applying Tasks and actions:
  - Installation and Configuration
  - Monitoring, Logging and Reporting
  - Changing
  - Supporting
  - Provisioning
  - Troubleshooting

Changes System State

Formal Definition of System Administration



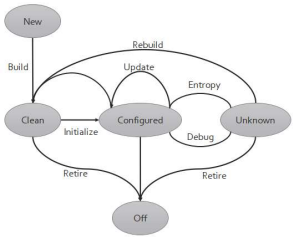
System State

- **System Policy**: specifications of a system's configuration and its acceptable usage.
- **System State S(t)**: the current configuration (files, kernel, memory or CPU usage) of a system.
- **Ideal State S*(t)**: States of the system that match the system policy. Over time, the system state shifts away from ideal state.
- **System Administration**: Modifying the system to bring it closer to S*(t).

- **Evard's Computer Machine Life Cycle:**
  - A Computer Machine Goes through different states in their life.
  - Different Processes are used for transition of the state of Computer Machines
  - Evard has outline different states and process that are performance on

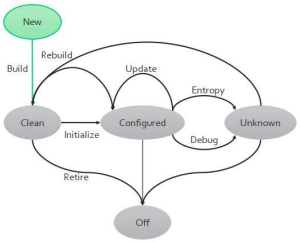
Evards ' Life Cycle Of Computer Machine

- Five States
  - New
  - Clean
  - Configured
  - Unknown
  - Off
- • Several Processes
  - Build
  - Initialize
  - Update
  - Entropy
  - Debug
  - Retire



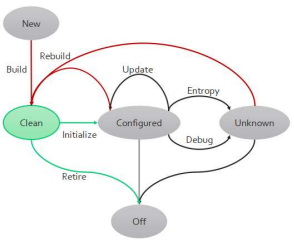
Computer Machine Life Cycle – New State

- **New State:**
  - A new machine
  - Unpacking
  - Physical Connections
  - Power up
  - Preinstalled OS or No OS



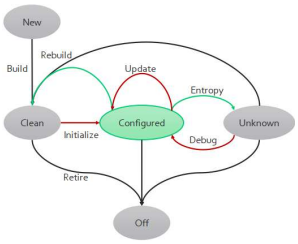
Computer Machine Life Cycle – Clean State

- **Clean State**
  - A machine that has the operating system installed but isn't set up to function in the System.
  - **Build Process**
    - Select OS
    - Install OS
    - Drivers and Software
  - **Rebuild Process**
    - Recovered Data
    - Reformat Disks
    - Reinstall OS/ Software



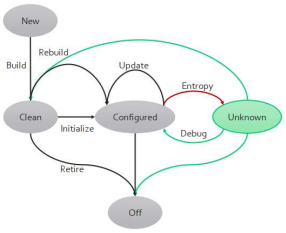
Computer Machine Life Cycle – Configured State

- **Configured State**
  - A machine that is properly configured to meet the needs of the System.
  - User Roles and Permissions
  - Updates
  - **Update Process**
    - Select Updates
    - Schedule Updates
    - Monitor



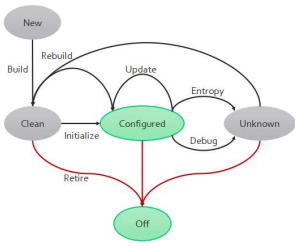
Computer Machine Life Cycle – Unknown State

- **Unknown State**
  - A machine that has been misconfigured, or that has become out of date, or that has been borrowed by an intern and returned stained
  - **Debug Process**
    - Troubleshooting
    - Remove the bugs
    - Fix Drivers and Software
  - **Entropy Process**
    - Changes by user/admin/updates
    - Compromised State
  - **Rebuild Process**
    - Recover User Data
    - Format Disks and Reinstall OS



Computer Machine Life Cycle – Off State

- **Off State**
  - Powered Off
  - No User Data
  - OS installed or Removed
  - Disconnected from Infrasture
  - **Retire Process**
    - Recover User Data
    - Flush the Hard Disks
    - Isolate from Infrastructure



System Administration Maturity Model (SAMM)

- Five SAMM levels:
  1. Ad Hoc
     - Ad-Hoc non-repeatable solutions, firefighting.
  2. Repeatable
     - Some repeatable process
  3. Defined
     - Documented standard process
  4. Managed
     - Process effectiveness measured, adapted.
  5. Optimized
     - Continuous Improvement

Level 1 Ad-Hoc – Common Practices

- **New user** – Verbal requests are addressed as time permits or escalated with management.
- **Software install** – New or upgraded software is installed whenever & where ever makes the most sense at the moment.
- **Hardware install** – New or upgraded hardware is installed whenever & where ever makes the most sense at the moment.
- **Problem report** – Problems are sometimes reported by users by mail or phone to a random administrator.
- **Security** – No specific security standards or policies exist.
- **Disk capacity** – Disk space is in short supply. No information on usage rates is available. Project managers of supported organizations often fight among themselves regarding disk space.
- **Backups** – Backups are usually done according to a weekly schedule.

Level 2 Repeatable – Common Practices

- **New user** – Procedure to request and create an account is well documented. Cycle time for requests is monitored.
- **Software install** – Installation guidelines are understood. Time spent installing and configuring software is tracked.
- **Hardware install** – Installation standards are understood. Time spent installing and configuring hardware is tracked.
- **Problem report** – Process to report problems is well understood by users and cycle time for problem resolution is monitored.
- **Security** – Various security standards are clearly documented. Security violations are monitored.
- **Disk capacity** – Acceptable disk capacity levels are established. Capacity is periodically monitored.
- **Backups** – Failure conditions for backups are understood. Failure rates and effort to resolve problems are tracked.

Level 3 Defined – Common Practices

- **New user** – Head count expansion information is provided to signal new account requests are expected. Revision of procedure.
- **Software install** – Installations are planned with the supported organizations. Reasons for install/upgrade are recorded.
- **Hardware install** – Installations are coordinated with supported organizations and vendors. Reasons for install/upgrade are recorded.
- **Problem report** – Problems are mapped to root causes. Group reviews solutions to resolutions suggested to address root causes.
- **Security** – Group reviews security incidents for root vulnerabilities. Resolutions are discussed, tested and released.
- **Disk capacity** – Capacity planning is addressed in project plans written by supported organizations and reviewed by the network & systems group.
- **Backups** – Training is provided for network and systems group to enhance backup programs and procedures.

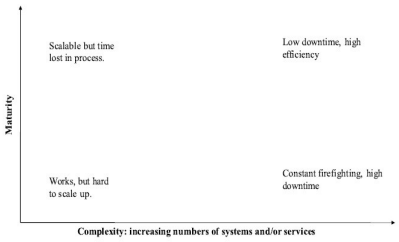
Level 4 Managed – Common Practices

- **New user** – Cycle time numbers are used to adjust staffing to meet demand and requirements.
- **Software install** – Productivity measures and goals for installation created.
- **Hardware install** – Productivity/problem data is compare to goals for installs, tests and demos.
- **Problem report** – Cycle time and resolution quality information is used on a regular basis to access effectiveness of problem reporting and resolution system.
- **Security** – Effectiveness of group reviews is studied.
- **Disk capacity** – Metric for disk availability is created and used.
- **Backups** – Backup system is certified with high reliability rating.

Level 5 Optimized – Common Practices

- **New user** – Accounts are electronically requested and verified.
- **Software install** – Problems with software installs are documented and avoidable with new procedures.
- **Hardware install** – New hardware technologies are evaluated & integrated.
- **Problem report** – New problem reporting system installed to better meet user requirements for ease of use.
- **Security** – Internal contest to establish better security practices is established.
- **Disk capacity** – Project tracking information combined with metrics of utilization are used to predict needs.
- **Backups** – Backup process is revisited with input from supported organizations re: production schedules.

Maturity and Complexity



Maturity Model Level Tools

- Ad-Hoc : OS GUI, CLI, or web administration interfaces
- Repeatable : Version Control (RCS, SVN, GIT), request tracker.
- Defined : Automatic monitoring (Nagios, Monit, god)
- Managed : Configuration management (AutomateIT, cfengine)
- Optimized: Evidence & observation recording, security checks integration (SAST/DAST)

Common Sys-Admin Tasks

1. Manage users (background stuff)
2. Manage hardware
3. Perform backups
4. Install new software systems
5. Troubleshooting
6. System monitoring
7. Auditing security
8. Help users
9. Communicate

User Management

- Creating user accounts
  - Consistency requires automation
  - Startup (dot) files
- Namespace Management
  - Usernames and UIDs
  - Multiple namespaces or SSI (Single Sign In)?
- Removing user accounts
  - Consistency require automation
  - Many accounts across different systems

25

Hardware Management

- Adding and removing hardware
  - Configuration, cabling, etc.
- Purchase
  - Evaluate and purchase servers + other hardware
- Capacity Planning
  - How many servers? How much bandwidth, storage?
- Data Center Management
  - Power, racks, environment (cooling, fire alarm)
- Virtualization
  - When can virtual servers be used vs. physical?

26

Backups

- Backup Strategy and policies
  - Scheduling: When and how often?
  - Capacity Planning
  - Location: on site vs offsite
- Monitoring backups
  - Checking logs
  - Verifying media
- Performing restores when requested.

27

Software Installation

- Automated consistent OS installs
  - Desktop vs Server OS image needs.
- Installation of Software
  - Purchase , find, or build custom software.
- Managing software installations
  - Distributing software to multiple hosts.
  - Managing multiple version of software package.
- Patching and updating software.

28

Troubleshooting

- Problem Identification
  - By user notification
  - By log files or monitoring programs
- Tracking and Visibility
  - Ensures users know you're working on problem
  - Provide an ETA if possible.
- Finding the root cause of problems
  - Provide temporary solution if necessary
  - Solve the root problem to permanently eliminate

29

System Monitoring

- Automatically monitor system for
  - Problems (disk full, error logs, security)
  - Performance (CPU, mem, disk, network)
- Provides data for capacity planning
  - Determine need for resources
  - Establish case to bring to management

30

Helping Users

- Request tracking system
  - Ensures that you don't forget problems.
  - Ensures users know you're working on their problem; reduce interruptions, status queries.
  - Lets management know what you've done.
- User documentation and training
  - Policies and procedures
- Schedule and communicate downtimes

Communicate

- Customer
  - Keep customer appraised of process.
    - When you've started working on a request with ETA.
    - When you make progress, need feedback.
    - When you're finished.
  - Communicate system status
    - Uptime, scheduled downtimes, failures.
  - Meet regularly with customer managers
- Managers
  - Meet regularly with your manager.
  - Write weekly status reports

Specialized Skills

- Heterogeneous Environments
  - Integrating multiple-OSes, hardware types, or network protocols, distributed sites.
- Database
  - SQL RDMS
- Networking
  - Complex routing , high speed networks, voice.
- Security
  - Firewalls, authentication, NIDS, cryptography
- Storage
  - NAS, SANS, cloud storage
- Virtualization and cloud computing
  - VMware, cloud architectures.

Qualities Of A Successful System-Administrator (Sys-Admin)

- Customer oriented
  - Ability to deal with interrupts, time pressure
  - Communication skills
  - Service provider, not system police
- Technical Knowledge
  - Hardware, network, and software knowledge
  - Debugging and troubleshooting skills
- Time Management
  - Automate everything possible
  - Ability to prioritize task: urgency and importance.

1st Step to Better System-Administrator

- Use request system
  - Customer know what you're doing.
  - You know what you're doing.
- Manage quick request right
  - Handle emergencies quickly
  - Use request system to avoid interruptions.
- Policies
  - How do people get help?
  - What is the scope of responsibility for SA teams?
  - What is our definition of emergency?
- Start every host in a known state.

Principles of A Good System-Administrator

- Simplicity
  - Choose the simplest solution that solves the entire problem.
  - Work towards a predictable system.
- Clarity
  - Choose a straightforward solution that's easy to change , maintain, debug, and explain to other SAs.
- Generality
  - Choose reusable solutions that scale up; use open protocols.
- Automation
  - Use software to replace human effort.
- Communication
  - Be sure that you're solving the right problems and that people know what you're doing.
- Basic First
  - Solve basic infrastructure problems before advances ones.

Professional System Administrator Organizations

- USENIX: Advance Computing Association
- LISA: Large Installation System Administration
- SAGE: System Administration Guild
- LOPSA: League of Professional System Administrator.

37

System Administrator Certifications

- CCNA, CCNP, CCIE (Cisco)
- cSAGE (SAGE)
- MCSA (Microsoft)
- RHCE (Red hat)
- SCSA (Sun)
- VCP (VMware)

38

Job Description Example - SAGE

- **Novice :** OS familiarity, help desk skills
- **Junior:** Can use system administration tools
- **Intermediate:** Understanding of distributed computing, common servers, automate small tasks, independent action
- **Senior:** Understanding of scaling issues, including capacity planning, solve problems by addressing the root cause, higher level programming abilities, write proposals for purchasing, data center planning, etc.

39

SAGE Code of Ethics

- Professionalism
- Personal Integrity
- Privacy
- Laws and Policies
- Communication
- System Integrity
- Education
- Social Responsibility

40

SYS-ADMIN Case STUDY: Terry Child

- Network Administrator for SAN Fransisco CCIE
  - who built city's FiberWAN network.
- Terry was only person with router passwords
  - IT department acknowledges knowing that
  - He was on-call 24x7x365 to resolve issues
- Terry refused to give password to boss
  - Cited fears that they would be misused by management, outside contractors.
- What was the right thing for Terry to do?

41

Other operational issues

1. An employee is working from home and has trouble joining a video call. How would you help them?
2. Some employees are complaining about their Internet speed. What steps would you take to identify the problem?
3. We would like to create simple manuals to help employees to use our equipment properly. What instructions would you give to help your colleagues use a printer?
4. How would you deal with a virus?

42

Key Points

- Defination
  - System, system state, ideal state, administration
- Principles of System Administration
  - Simplicity
  - Clarity
  - Generality
  - Automation
  - Communication
  - Basic First
- System Administration Maturity Model
  - Maturity and Complexity, tools

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Lec01b - Hardware

Lecture Topics

1. Servers vs Desktops
2. Server Hardware
3. Different Approaches to Servers
4. Server Selection Options & Considerations

What are Servers?

- A group of computer machines in an organization
- Used to provide different types of services:
  - Application services
  - Web services
  - Back-office processing
  - Databases
  - Batch computation
  - Etc.

Servers vs Services

- A server is a computer machine consisting of
  - Hardware
  - Operating System
  - Software packages

Server Hardware

- CPU
- Memory
- Storage
- Power Supply
- Network Interface
- Motherboard
- Peripheral Devices
- Rack

Server OSes

- Linux
- Windows Server
- UNIX

Server Software Packages

- Database
- Email
- Network

How are Servers Different?

- 1000s of clients depend on server.
- Requires high reliability.
- Requires tighter security.
- Often expected to last longer.
- Investment amortized over many clients, longer lifetime.

Product Lines

- Home
  - Cheapest purchase price
  - Components change
- Business
  - Focuses on Total Cost of Ownership (TCO)
  - Slower hardware changes, longer lifetime
- Server
  - Lowest cost per performance metric (nfs, web)
  - Easy to service rack-mountable chassis
  - High quality (MIL-SPEC) components

Key Types of Servers by Hardware

- Tower Servers: Housed in upright, standalone cabinets (similar to desktop PCs), these are ideal for small businesses needing basic, cost-effective, and easy-to-cool systems.
- Rack Servers: Designed to be installed into standardized 19-inch metal frames (racks), these are general-purpose, space-efficient, and easy to scale, making them ideal for data centers.
- Blade Servers: Modular, ultra-thin units (blades) that slide into a chassis, sharing power and cooling. They provide the highest density for processing power in limited space.
- Mainframe Servers: Large, highly robust computers designed for high-volume, secure, and mission-critical processing, often used by large enterprises.
- Micro Servers: Compact, low-power servers optimized for specific, light workloads.

Key Types of Servers by Hardware

- Micro Server

- Tower Server

- Rack Server

- Blade Server

- Mainframe Servei

- Supercomputer

Common Types of Servers by Function

- **Web Server:** Hosts websites and delivers web pages (e.g., Apache, Nginx).
- **Database Server:** Stores, manages, and retrieves large datasets (e.g., MySQL, Oracle).
- **Mail Server:** Handles sending, receiving, and storing emails (e.g., Exchange, Gmail).
- **File Server:** Centralized storage for file sharing and access.
- **Application Server:** Runs applications for users over a network.
- **DNS Server:** Translates domain names (like google.com) into IP addresses.
- **Proxy Server:** Acts as an intermediary, enhancing security and performance.
- **FTP Server:** Facilitates file transfers using the File Transfer Protocol.
- **Print Server:** Manages print jobs for multiple users on shared printers.
- **DHCP Server:** Automatically assigns IP addresses to devices.
- **Gaming Server:** Hosts multiplayer online games.
- **Cloud Server:** Virtualized resources accessed over the internet.
- **Virtual Server:** Multiple virtual servers on one physical machine.

Why Setup a Server?

- More internal space
- More CPU and/or memory
  - More higher-end CPUs
  - More faster memory
- High performance I/O
  - PCIe vs PCI
  - SCSI/FC-AL vs IDE
- Rack-mounted
- Redundancy
  - RAID
  - Hot-swap, Hot-spares

Rack Mounting

- Efficient space utilization
  - Simple. Rectangular shape measured in Rus
  - Repair and upgrade while mounted in rack
  - No side access required
- Requirements
  - Cooling through back, not sides
  - Drives in front, cables in back
  - Remote management (serial console, hardware sensors, VM MUI)

Server CPUs

- There are two main types of server processors:
- X86
  - X86 processors are the most common type of processor found in servers.
  - They are made by companies such as Intel and AMD.
  - X86 processors are designed for general-purpose computing.
  - They can be used for a variety of tasks, including web hosting, database management, and file sharing.
- RISC
  - RISC processors are designed for specific tasks.
  - They are often used in high-performance servers.
  - RISC processors are made by companies such as IBM and Oracle.

Selection of CPU

- The type of server processor you need depends on the type of server you are using.
- If you are using a **general-purpose server**, an x86 processor is likely the best choice.
- If you are using a **high-performance server**, a RISC processor may be the better choice.

Other CPU Considerations (1)

- **Clock speed:**
  - Keep up with the demands of modern businesses
  - Ablity to process large amounts of data quickly and efficiently
- **Cores:**
  - • number of cores in a processor can have a big impact on its performance.
  - • More cores means that the processor can handle more tasks at the same time.
  - • in the market today have up to 32 cores and more
- **Memory support:**
  - Support large amounts of memory
  - Need to store and process large amounts of data
- **Expand-ability:**
  - Need to be expandable so that businesses can add more features as their needs change.
  - Built-in features such as security or management tools
  - Expansion slots so that businesses can add more features as they need them

Other CPU Considerations (2)

- **Efficient Data Management:**
  - Data management is a key concern
  - Must be able to efficiently handle large amounts of data
  - Keep the server running smoothly, even when under heavy load
- **Cost and Power Consumption:**
  - Most important factors to consider is cost
    - Initial Cost
    - Operational Cost
  - Energy-efficiency
    - Right balance between power consumption and performance to minimize your carbon footprint
- **Budget:**
  - key factors to consider is your Budget
- **Workload:**
  - The server load

Server CPU Choices

1. Intel Xeon
   - Multicore with two threads per core, 1.8 to 3.3 Ghz, 8 cores
   - Up to 18 MB L3 Cache
2. AMD Opteron
   - 4, 6, 8, or 12 cores @ 1.4 to 3.2 GHz
   - Up to 12 MB L3 cache
3. IBM Power 7/8/9/10…
   - 4, 6, or 8 cores with 4 threads each @ 3.0 to 4.25 GHz
   - 4 MB L3 cache per core (up to 32MB for 8-core)
4. Sun Niagara 3
   - 16 cores with 8 threads each @ 1.67 GHz
   - 6 MB L2 cache

Xeon vs Pentium/Core CPUs

- Xeon based on Pentium/Core with changes that vary by model:
  - Allows more CPUs
  - Has more cores
  - Better hyper-threading
  - Faster/larger CPU caches
  - Faster/larger RAM support

Intel Processor Families

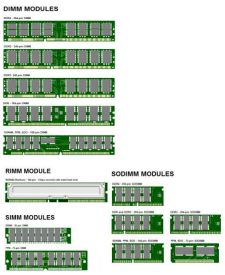
1. Intel® Xeon® Scalable Processors
   - 4th Gen Intel® Xeon® Scalable processors have the most built-in accelerators of any CPU on the market to
   - improve performance in AI, analytics, networking, storage, and HPC.
2. Intel® Xeon® Max Series
   - Maximize bandwidth with the Intel® Xeon® CPU Max Series, the first and only x86-based processor with high-
   - bandwidth memory (HBM).
3. Intel® Xeon® W Processor
   - Designed for creative professionals, delivering the performance you need for VFX, 3D rendering, and 3D CAD on a workstation.
4. Intel® Xeon® D Processor
   - When space and power are at a premium, these innovative system-on-a-chip processors bring workload
   - optimized performance.
5. Intel® Xeon® E Processor
   - Essential, business-ready performance, expandability and reliability for entry server solutions.

Server Memory

- Server memories are concerned with
  - RAM
  - CPU Cache
- Two Major aspects of server memory are
  - Large Capacity
  - Higher speed
- x86 supports up to 64GB with PAE
- x86-64 supports 1 PB (1024 TB)
- Servers need faster RAM than desktops
  - Higher memory speeds.
  - Multiple DIMMs accessed in parallel.
  - Larger CPU caches.

Types of Memory

- FBDIMM
  - Fully buffered
- UDIMM
  - Unbuffered dual in-line memory module
- RDIMM
  - Registered dual in-line memory module
- LRDIMM
  - Load Reduced dual in-line memory module
- SODIMM
  - Used with laptops
- MicroDIMM



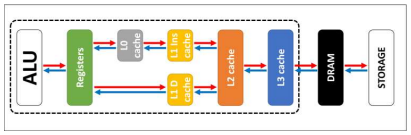
Transfer Speed of RAM

- SDRAM
  - Synchronous Dynamic Random Access Memory
- DDR
  - Double Data Rate SDRAM
- DDR2
  - 2 x times the DDR
- DDR3
  - low power, Twice clock multiplier with a four times clock multiplier ,
  - No F/B Compatibility, 2 x DDR2 TT
- DDR4
  - low power higher module density and lower voltage
  - requirements, coupled with higher data rate transfer speeds

DDR SDRAM Standard	Internal rate (MHz)	Bus clock (MHz)	Prefetch	Data rate (MT/s)	Transfer rate (GB/s)	Voltage (V)
SDRAM	100-166	100-166	1n	100-166	0.8-1.3	3.3
DDR	133-200	133-200	2n	266-400	2.1-3.2	2.5/2.6
DDR2	133-200	266-400	4n	533-800	4.2-6.4	1.8
DDR3	133-200	533-800	8n	1066-1600	8.5-14.9	1.35/1.5
DDR4	133-200	1066-1600	8n	2133-3200	17-21.3	1.2

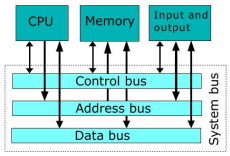
The L1, L2, L3 Cache

- Cache memory is high speed memories placed between RAM and CPU.
  - RAM -> L3 -> L2 – L1-> Registers - > CPU



System Bus

- Data sharing
- Addressing
- Power
- Timing

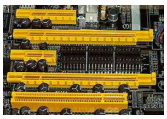


Expansion Bus Types

- Common expansion bus types are:
  - ISA- Industry Standard Architecture:
    - Designed for use in the original IBM PC
    - 8 bit / 16 bit
    - The ISA bus ran at a clock speed of 4.77 MHz and improved version 8MHz.
    - 16-bit version of the ISA bus is sometimes known as the AT bus (AT-Advanced Technology).
  - MCA - Micro Channel Architecture:
    - IBM developed this bus as a replacement for ISA when they designed the PS/2 PC in 1987
    - Speed of 10MHz and supported either 16-bit or 32-bit data
  - EISA - Extended Industry Standard Architecture
    - As an alternative to MCA
    - Use a 32-bit data path and provided 32 address lines, giving access to 4GB of memory
    - Ran at 8MHz in order for it to be compatible with ISA
  - VESA - Video Electronics Standards Association
    - invented to help standardize PCs video specifications
    - 32-bit data path and ran at 25 or 33 MHZ
    - VL-Bus was superseded by PC
  - PCMCIA - Personal Computer Memory Card Industry Association (Also called PC bus)
  - AGP - Accelerated Graphics Port
  - SCSI - Small Computer Systems Interface
  - Universal Serial Bus (USB)

Expansion Bus Types

- Servers need high I/O throughput.
  - Fast peripherals: SCSI-3, Gigabit ethernet
  - Often use multiple and/or faster buses.
- PCI
  - Desktop: 32-bit 33 MHz, 133 MB/s
  - Server: 64-bit 66 MHz, 533 MB/s
- PCI-X (backward compatible)
  - v1.0: 64-bit 133 MHz, 1.06 GB/s
  - v2.0: 64-bit 533 MHz, 4.3 GB/s
- PCI Express (PCIe)
  - Serial architecture, v3.0 up to 16 GB/s



Server Power Supply

- Servers based on the ATX or microATX form factors generally use an ATX power supply
- Pedestal servers based on one of the Server System Infrastructure (SSI) form factors generally use an EPS12V power supply

ATX Power Supply Standards

- ATX power supplies were originally designed for use in desktop computers.
- Widely used in entry-level tower and slimline servers.
- ATX power supply standards include the following:
  - ATX version 2.03 Older entry-level tower servers
  - ATX12V More recent entry-level tower servers
  - ATX1U 1U slimline servers
  - ATX2U 2U slimline servers

ATX1U/2U Rack-Mounted Power Supplies

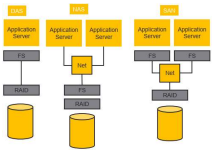
- Rack-mounted servers use a variety of power supply standards.
- Most 1U and 2U servers use the power supply standards developed by the SSI Forum, such as the ATX1U and ATX2U power supplies
- The ATX1U and ATX2U power supply standards use the same 20-pin ATX power supply, floppy, and hard disk power connectors used by the ATX 2.03 and ATX12V v1.x power supply standards.
- Some 200w and larger ATX1U power supplies also feature the 4-pin ATX12V power supply connector
- ATX2U power supplies feature the 6-pin auxiliary power supply connector

SSI Power Supplies

- The SSI Forum has developed a series of power supply and connector form factors designed for use in various types of servers.
- Pedestal-mounted servers
  - EPS12V
  - ERP12V
- 1U rack-mounted servers
  - EPS1U
- ➢2U rack-mounted servers
  - EPS2U
  - ERP2U

Storage Types

- Four Types of Storage Solutions are:
  - Direct-attached storage (DAS)
    - Connected with SAS, SATA or
    - PCIe
  - Network-attached storage (NAS)
  - Storage area network (SAN)
  - Cloud storage
    - Requires Internet



Storage Types

- **Direct-attached storage**
  - HDD
    - Applications : Suitable for Write Heavy Applications, Backups
    - Advantages : Cheap, High Capacity, Unlimited number of writes , Low Cost
    - Disadvantages : Frgile, Breaks down over time, Slow
  - SSD
    - Applications : General storage , Boot device
    - Advantages : Relatively cheap , Fast enough for most , Enough capacity for most
    - Disadvantages : Limited number of writes , May not be suitable for enterprise use
  - NVME SSD
    - Applications : High-performance computing , Boot drives , Databases
    - Advantages : Currently the fastest persistent server storage type on the market
    - Disadvantages : Can be very expensive , Limited number of writes

Storage Types

- **Network-attached storage**
  - Applications : Provides access to the same data on multiple systems
  - Advantages : Can be very fast and very high capacity, Can mix SSDs and hard drives to make a faster solution than hard drives alone, RAID storage always
  - Disadvantages : Requires a separate computer, Expensive
- **Storage area network**
  - Applications : Sophisticated databases , Virtualization deployments , Large virtual desktop infrastructures (VDIs) , Enterprise resource management workloads
  - Advantages : Resilient, Removes single points of failure , Easily withstands failure of multiple components/devices
  - Disadvantages : Expensive , Complex
- **Cloud Storage**
  - Applications : Off-site backups , Provides access to data from multiple locations
  - Advantages :Peace of mind
  - Disadvantages : Requires Internet connection , Usage rates may impact upload/download capacity

SAS , SATA SSDs and NVME

- **Hard Disks:**
  - Three RPM speeds: 7,200 RPM, 10,000 RPM, and 15,000 RPM
  - Capacity varies, starting at about 300GB and going up to over 10TB
  - SATA 6, SAS 6, and SAS 12 are typical connection speeds
- **SSDs:**
  - NAND flash memory cells and are always faster than hard drives, primarily
  - because they do not have to "seek" the data on the disk, and latency is low
- **SAS or SATA connector:**
  - SATA 6, SAS 6, and SAS 12, SAS 24
  - High Cost
  - SSDs wear out as they are written to
  - SAS and SATA bus interfaces were originally designed for slow, mechanical hard drives
- **NVMe:**
  - NVMe storage can connect via a wide range of connector types from M.2 to U.2 to U.3 to newer standards such as E1.S, also known as "ruler" SSDs
  - Connect over the PCIe bus, typically an x4 connection, though x2 and x8 are also possible
  - The maximum speeds for Gen 3 PCIe and Gen 4 PCIe are 3.5GB/s and 7 GB/s, respectively

RAID and its Levels

- RAID (redundant array of independent disks)
- A way of storing the same data in different places on multiple hard disks or solid-state drives (SSDs) to protect data in the case of a drive failure.
- There are different RAID levels, however, and not all have the goal of providing redundancy.

RAID 0

RAID 1

Raid 5

RAID 6

RAID 10 (RAID 1 + 0)

RAID 50 (RAID 5 + 0)

RAID 60 (RAID 6 + 0)

Server Racks

- Racks organize IT equipment into standardized assemblies that make efficient use of space and other resources
- At the most basic level, a rack consists of two or four vertical mounting rails and the supporting framework required to keep the rails in place.
- The rails and framework are typically made of steel or aluminum to support hundreds or even thousands of pounds of equipment.
- The width of the rails, the horizontal and vertical spacing of the mounting holes, the size of the equipment cabinets and other measurements are standardized
- Rack Standardization
  - Most IT equipment is nominally 19 inches wide (including mounting hardware) and follows a standard set by the Electronics Industry Alliance (EIA) and now maintained by the Electronic Components Industry Association (ECIA).
  - The current 19-inch rack standard is called EIA-310-E, which is essentially equivalent to IEC-60297-3-100 or DIN 41494 in other regions. (There's also a standard for 23-inch wide telecom equipment.

Rack Units

- Although 19-inch racks are always the same nominal width, the height and depth vary.
- The depth of the rack rails is usually adjustable to some degree.
- The height of the rack is divided into standardized segments called rack units. Each rack unit is 1.75 inches high, and the height of a rack or an equipment cabinet is expressed as the number of rack units followed by the letter "U". For example, a 42U rack contains 42 rack units.
- That does not mean the rack is exactly 42 x 1.75 inches high because racks usually include at least a little extra space at the top and bottom that isn't usable rack space.
- It does mean that the rack will accommodate any combination of standard rack equipment up to 42U – whether it's 42 x 1U switches, 14 x 3U servers or 21 x 1U switches with 7 x 3U servers.
- Remember that the rack also has to be deep enough for the equipment and rated to support the combined weight of all the equipment.

Rack Types

- Open frame racks are just that—open frames with mounting rails, but without sides or doors.
- Rack enclosures have removable front and rear doors, removable side panels and four adjustable vertical mounting rails (posts).
- Wall-mount racks are designed to be attached to the wall, saving floor space and fitting in areas where other racks can't. They can be open frame racks or enclosed cabinets.

Rack Choices

- Basic rack options:
  - Doors
  - Side Panels
  - Roof
  - Casters and Levelers
  - Locks
  - Hinged Wall Bracket
  - Mounting Holes
  - Color
  - Toolless Mounting
- Rack enclosure and cooling:
  - Basic Airflow
  - Side Panels
  - Airflow Management
  - Cable Management
  - Thermal Ducts
  - Active Heat Removal
  - Close-Coupled Cooling
- Factors influencing Rack Choice:
  - Airflow and cooling
  - Equipment width
  - Security options
- Other Considerations
  - Power Distribution
  - Battery Backup
  - Device Management
  - Patch Panels
  - Environmental Monitoring
  - Security
  - Shock Pallet
  - Knockdown
  - Stability
  - Environmental Protection
  - Seismic Protection

Other Server Selection Factors

- Extensibility
- More CPU performance
- High-performance I/O rack mountable
- Upgrade options
- No side-access needs
- High-availability options
- Maintenance contracts
- Management options

Server Architecture / Approaches

1. **Primary Approaches**:
   - **All eggs in one basket:**
     - One machine used for many purposes
   - **Beautiful snowflakes:**
     - Many machines, each uniquely configured
   - **Buy in bulk, allocate fractions:**
     - Large machines partitioned into many smaller virtual machines using virtualization or containers
2. **Other Approaches:**
   - **Grid computing:**
     - Many machines managed one as unit
   - **Blade servers:**
     - A hardware architecture that places many machines in one chassis
   - **Cloud-based compute services:**
     - Renting use of someone else's servers
   - **Software as a service (SaaS):**
     - Web-hosted applications
   - **Server appliances:**
     - Purpose-built devices, each providing a different service

Server Reliability

- Levels of Redundancy
  - N+0, N+1,…, N+r
  - Identify Redundant Units, Power Supplies, CPUs, Hard disks etc
- Data Integrity
  - Ensure data is available and valid in future
  - RAID , Service Failure Vs Component failure
  - Non-RAID Approaches,
    - Redundant Systems vs Redundant Components
    - Distributed systems e.g Google's GFS, Hadoop's HDFS, Cassandra
- Hot-Swap Components
  - The ability to add, remove, and replace a component while the system is running
  - Not All components are hot swappable
- Servers Should Be in Computer Rooms or Data Centres

Remote vs Central Management

- Services should be remotely managed. If you have to be on the console of a machine to do rudimentary maintenance and configuration, each task will be more time consuming.
  - This is also an indication that automating management tasks may not be possible.
- Two basic reasons for remote servers
  - **Specialized Premises**
    - Servers are generally located in machine rooms where power and cooling can be managed more effectively. Remote management permits servers to be located anywhere. It is less expensive to rent colocation space elsewhere than to build a computer room in your office. It is often also more reliable.
  - **Geographic Redundancy**
    - Services that are replicated in many places must be manageable in a centralized fashion. Remote management opens up the possibility of deploying instances of the service around the world. Configuration changes and upgrades should not require manually updating each instance.
- Consider high latency over long-distance connections
  - Protocols may time out. User interfaces may work but can be unusably slow. This is not a limit of the technology, but rather bad software engineering. Plenty of APIs and user interfaces work well over high-latency links—and so should management tools.

Remotely Managing Servers

- Do all system administration tasks involving the machine from a remote location, except physical labor such as adding and removing physical hardware.
  - Ability to remotely access the machine's console and, optionally, have remote control of the power switch.
  - Ability to operate a system's console when the machine is in a bad state or otherwise disconnected from the network. e.g. Accessing a basic BIOS configuration, CTRL-ALT-DEL keys
- Saves time and cost
- Need to consider security aspects
- Two Types of Remote management techniques:
  - Integrated Out-of-Band Management
  - Non-integrated Out-of-Band Management

Integrated Out-of-Band Management

- Integrated Out-of-Band Management
  - Modern servers have remote management capabilities built-in. Such systems are generically called out-of-band (OOB) management and have an Ethernet port as an interface.
- OOB interface requires an IP Address, and remote system console is accessed via browser or client.
  - Remote access via a browser has limited functionalities as compared to clients.
- The remote management systems must be isolated from the main network and must be accessible even when the main system is down or powered off
- Security Aspects
  - Built-in systems have been found to suffer from traditional security holes, such as buffer overruns
  - Don't assume that a password on the OOB interface and using SSL is sufficient protection
  - OOB interface is equivalent to physical access to the machine
  - Put OOB interfaces on a dedicated protected network, use a web proxy, authentication and authorizatio

Non-Integrated Out-of-Band Management

- The built in OOB has some limitations in terms of remote power cycling and console access. Third party tools can address these issues.
  - **Remote Power Cycle:**
    - Switched power distribution unit (PDU) provide the remote power cycling.
    - A PDU is the fancy name for a power strip
  - **Remote Console with IP-KVM:**
    - Keyboard, video screen, and mouse (KVM) switch is a device that allows many machines share a single set of I/O peripheral devices.
    - An IP-KVM switch provides the remote console access.
    - An IP-KVM is a KVM switch that can be accessed remotely. This eliminates the need for any monitors, keyboards, or mice in the computer room. You simply run a client on your workstation that connects to the IP-KVM.
  - **Remote Console with Serial Consoles:**
    - Network equipment, appliances, and older servers have serial consoles. They have no video, keyboard, or mouse connector.
    - In the old days one would attach a physical VT-100 terminal to the serial console port of each device
    - In the 1990s it became popular to replace physical terminals with a terminal console concentrator.
    - Also, serial console concentrators usually have the option to require authentication—for example, using RADIUS or TACACS—before allowing someone to connect to a console.

Separate Administrative Networks

- It is common for servers to have a separate NIC that is connected to a dedicated administrative network.
- Separating administrative traffic from normal service traffic has a number of benefits:
  - For servers, the primary benefit is often to isolate disruptive traffic.
  - Backups, for example, consume a large amount of bandwidth and can interfere with normal service traffic.
    - Sometimes servers have a dedicated backup interface that uses a high-speed network dedicated to backups.
- ➢ Characteristics of administrative networks

Stable & Static	Strict Security	Dedicated	Separation	Simplicity	Availability
• Should be more stable • More static • Service Networks are more dynamic	• More Restrictive firewall rules • Serves more critical systems	• Only for administrative purpose	• Should be separated from other service network	• Simple Engineering • No VLANs • Simple Topology	• Should be always available • Immune to outages

Maintenance Contracts And Spare Parts (1)

- **Vendor SLA**
  - There is a variety of maintenance contract options, with different service level agreements (SLAs).
    - Replacement for a bad part with a 4-hour response time, a 12-hour response time, or next-day options.
      - Sometimes the options include 24/7 response, but other times just specify some number of business day
    - Other options include having the customer purchase a kit of spare parts and receive replacements after a spare part gets used.
      - Usually, the installation of the replacement part is by organization itself, though vendors offer onsite repairs for a higher price. Sometimes the service offerings vary by country or region, depending on where the vendor has local presence.

• **Response time**	• **Critical Servers**	• **Non-Critical Servers**
❑ Response Start ❑ Fast Response Vs Cost ❑ Plan for redundancy	❑ Lowest level maintenance contract ❑ Redundant Server ❑ Maintenance contract with warranties ❑ Vendors to maintain spares in same city	❑ Default maintenance contract ❑ Flexible Response time ❑ Low cost maintenance contract

Maintenance Contracts And Spare Parts (2)

- **Spare Parts**
  - In house repair department vs maintenance contract
    - Cost
    - Trainings, Certifications on the hardware
    - Maintaining inventory of spares
  - Self-Support Plans
    - Repair kit with maintenance contract
    - Homogeneous Infrastructure to allow sharable spares
    - No Licenses
    - Cold Spares
    - Suitable for High End Servers
  - Cross-Shipping
    - A type of shipping in which vendors requires the faulty units to be received before shipping replacement unit.
      - Alternate approach may be that vendor ship the replacement immediately and faulty unit is sent back afterwards.
      - Vendors use it to avoid unnecessary replacements
    - Cross shipping should be part of maintenance contract

Maintenance Contracts And Spare Parts (3)

- **Tracking Service Contracts**
  - Machines not covered in maintenance contract
    - Discovered during outage
    - Generally , coordinating with sales staff of vendor will add to maintenance contract
  - **Improvement Strategies**
    - Service contracts with 10% additional price
      - allows adding new machines as added
      - vendors can easily extend maintenance contract
    - Ensure each machine is purchased with maintenance contract
      - Avoid situations where vendor can agree to add new machines into existing maintenance contracts
    - Tract expiry of the initial maintenance contract that came with purchase
      - Remove dismantled machines
      - Add new machines

Selecting Vendors With Server Experience

- Some vendors have years of experience designing servers, and it shows
- They build hardware with the common server-related features, as well as include little extras that one can learn only from years of market experience.
- Servers from inexperienced vendors may lead to…
  - Issues to integrate with enterprise systems such as authentical systems
  - Management of the device is manual and cumbersome, lacking the ability to manage many servers at once, and with no kind of remote management
  - No maintenance contracts or degraded maintenance contracts
  - Low quality technical support
  - Issues in replacement of parts
- Select vendors that are known for building reliable hardware. Some vendors cut corners by using consumer-grade parts; others use premium-quality parts.

1

DSM2294 – Lecture 3a

LINUX FILE MANAGEMENT & COMMAND LINE INTERFACE CLI

2

Linux File Management

- The most frequent tasks that you perform on your PC is creating, moving or deleting Files. Let's look at various options for File Management.
- To manage your files, you can either use
  1. Terminal (Command Line Interface - CLI)
  2. File manager (Graphical User Interface -GUI)
- For this course, we will be working on the Linux using the Terminal.

3

Why learn Command Line Interface?

- Even though the world is moving to GUI based systems, CLI has its specific uses and is widely used in scripting and server administration.
- Let's look at some of CLI's compelling uses -
  - Comparatively, CLI offer more options & are flexible. Piping and stdin/stdout are immensely powerful are not available in GUI
  - Some configurations in GUI are up to 5 screens deep while in a CLI it's just a single command
  - Moving, renaming 1000's of the file in GUI will be time-consuming (Using Control /Shift to select multiple files), while in CLI, using regular expressions so can do the same task with a single command.
  - CLI load fast and do not consume RAM compared to GUI. In crunch scenarios this matters.
- **Both GUI and CLI have their specific uses.**
  - For example, **in GUI, performance monitoring graphs** give **instant visual feedback** on system health, while seeing hundreds of lines of logs in CLI is an eyesore.

4

Launching the CLI on Ubuntu (1)

- There are 2 ways to launch the terminal (CLI):
  1. Go to the Dash and type terminal
  2. Or you can press **CTRL + Alt + T** to launch the Terminal

5

Launching the CLI on Ubuntu (2)

- Once you launch the CLI (Terminal), you would find something as username@VirtualBox written on it.
  1. The first part of this line is the name of the **user** (bob, tom, ubuntu, home...)
  2. The second part is the computer name or the host name. The hostname helps identify a computer over the network. In a server environment, host-name becomes important.
  3. The ':' is a simple separator
  4. The tilde '~' sign shows that the user in working in the **home directory**. If you change the directory, this sign will vanish.
  5. The '$' sign suggests that you are working as a regular user in Linux. While working as a root user, '#' is displayed.

6

Present Working Directory

- The directory that you are currently browsing is called the **Present Working Directory**.
- You log on to the home directory when you boot your PC.
- If you want to determine the directory you are presently working on, use the command: **pwd**
  - PWD command stands for **P**rint **W**orking **D**irectory

Changing Directories

7

- If you want to change your current directory use the '**cd**' command.
- Consider the following example:

```
user@VirtualBox: ~$ cd /tmp
user@VirtualBox: /tmp$ cd /bin
user@VirtualBox: /bin$ cd /usr
user@VirtualBox: /usr$ cd /tmp
user@VirtualBox: /tmp$
```

- Here, we moved from directory /tmp to /bin to /usr and then back to /tmp.

Navigating To Home Directory

8

- If you want to navigate to the home directory, then type **cd**:

```
user@VirtualBox: ~/Documents/test$ cd
user@VirtualBox: ~$
```

- You can also use the **cd ~** command:

```
user@VirtualBox: ~/Documents/test$ cd ~
user@VirtualBox: ~$
```

Moving To Root Directory

9

- The root of the file system in Linux is denoted by '/'.
- This is similar to 'c:\' in Windows.
- Note: In Windows, you use backward slash "\" while in UNIX/Linux, forward slash "/" is used
- Type 'cd /' to move to the root directory.

```
user@VirtualBox: ~$ cd /
user@VirtualBox: /$
```

- **TIP**: Do not forget space between **cd** and **/**. Otherwise, you will get an error.

Navigating Through Multiple Directories

10

- You can navigate through multiple directories at the same time by specifying its complete path.
  - Example: If you want to move the /cpu directory under /dev, we do not need to break this operation in two parts.
- Instead, we can type '/dev/cpu' to reach the directory directly.

```
user@VirtualBox: ~$ cd /dev/cpu
user@VirtualBox: /dev/cpu$
```

Moving Up One Directory Level

11

- For navigating up one directory level, try **cd ..**

```
user@VirtualBox: /dev/cpu$ cd ..
user@VirtualBox: /dev$
```

- Here by using the 'cd ..' command, we have moved up one directory from '/dev/cpu' to '/dev'.
- Then by again using the same command, we have jumped from '/dev' to '/' root directory.

Relative and Absolute Paths

12

- A path in computing is the address of a file or folder.
- Example:
  - In Windows: **C:\documentsandsettings\user\downloads**
  - In Linux: **/home/user/downloads**
- There are two kinds of paths:
  1. Absolute Path
  2. Relative Path

Absolute Path

13

- Let's say you have to browse the images stored in the Pictures directory of the home folder
- The absolute file path of Pictures directory: **/home/username/Pictures**
- To navigate to this directory, you can use the ***cd /home/username/Pictures*** command:

```
user@VirtualBox: ~$ cd /home/username/Pictures
user@VirtualBox: ~/Pictures$
```

- This is called absolute path as you are specifying the full path to reach the file.

Relative Path

14

- The Relative path comes in handy when you have to browse another subdirectory within a given directory.
  - It saves you from the effort to type complete paths all the time.
- Suppose you are currently in your Home directory, and you want to navigate to the Downloads directory.
- You do no need to type the absolute path

```
user@VirtualBox: ~$ cd /home/username/Downloads
user@VirtualBox: ~/Downloads$
```

- Instead, you can simply type **'cd Downloads'** and you would navigate to the Downloads directory as you are already present within the '**/home/username**' directory.

```
user@VirtualBox: ~$ cd Downloads
user@VirtualBox: ~/Downloads$
```

- This way you do not have to specify the complete path to reach a specific location within the same directory in the file system.

Summary

15

- To manage your files, you can use either the GUI(File manager) or the CLI(Terminal) in Linux.
  - Both have its relative advantages but for this course, we will focus on the CLI aka the Terminal
- You can launch the terminal from the dashboard or use the shortcut key **Cntrl + Alt + T**
- The pwd command gives the present working directory.
- You can use the cd command to change directories
- Absolute path is complete address of a file or directory
- Relative path is relative location of a file of directory with respect to current directory
- Relative path help avoid typing complete paths all the time.

1

DSM2294 – Lecture 3b

BASIC LINUX COMMANDS

2

Listing files (ls)

- If you want to see the list of files on your Linux system, use the '**ls**' command.
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    - Directories are denoted in blue color.
    - Files are denoted in white.
    - You will find similar color schemes in different flavors of Linux.
- You can use '**ls -R**' to shows all the files not only in directories but also subdirectories
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7th Column	Directory or File name

4

Listing Hidden Files

- Hidden items in Linux begin with a "." (period symbol) at the start, of the file or directory.
- Any Directory/file starting with a '.' will not be seen unless you request for it. To view hidden files, use the command: `ls -a`

5

Creating & Viewing Files (1)

- The 'cat' command is used to display text files. It can also be used for copying, combining and creating new text files.
- Let's see how it works.
- To create a new file, use the command:
  - **cat >** filename
  - Add content
  - Press 'ctrl + d' to return to command prompt.
- To view a file, use the command: `cat filename`

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Creating & Viewing Files (2)

7

- Let's see the file we just created:

```
user@VirtualBox:~$ cat sample1
This is sample1
```

- Let's see another file (sample2):

```
user@VirtualBox:~$ cat > sample2
This is sample2
```

- The syntax to combine 2 files is:

```
cat file1 file2 > newfilename
```

Creating & Viewing Files (3)

8

- Let's combine sample 1 and sample 2:

```
user@VirtualBox:~$ cat sample1 sample2 > sample
```

- As soon as you insert this command and hit enter, the files are concatenated, but you do not see a result.
  - This is because the Bash Shell (Terminal) is a **silent type**.
  - It will never give you a confirmation message like "OK" or "Command Successfully Executed".
  - It will only show a message when something goes wrong or when an error has occurred.
- To view the new combo file "sample" use the command: `cat sample`
- **Note:** Only text files can be displayed and combined using this command.

Deleting Files

9

- The 'rm' command removes files from the system without confirmation.
- To remove a file use the syntax: `rm filename`

Moving and Re-naming files (1)

10

- To move a file, use the command: `mv filename new_file_location`
- Suppose we want to move the file "sample2" to location /home/user/Documents.
  - Executing the command: `mv sample2 /home/user/Documents`
  - Note: the mv command needs super user permission. If you are executing the command as a standard user, you will get an error.
  - To overcome the error use command: `sudo command_you_want_to_execute`

Moving and Re-naming files (2)

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- After authentication, the system will invoke the requested command.
- Sudo maintains a log of each command run.
- System administrators can trackback the person responsible for undesirable changes in the system, e.g.

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[sudo] password for user: ****
user@VirtualBox:~$
```

Moving and Re-naming files (3)

12

- For renaming file(s): `mv filename newfilename`
  - **NOTE**: By default, the password you entered for sudo is retained for 15 minutes per terminal. This eliminates the need of entering the password time and again.
- You only need root/sudo privileges if the command involves files or directories not owned by the user or group running the commands

Creating Directories

13

- Directories can be created on a Linux operating system using the following command: `mkdir directoryname`
- This command will create a subdirectory in your present working directory, which is usually your "Home Directory".
  - For example: `mkdir mydirectory`
- If you want to create a directory in a different location other than 'Home directory', you could use the following command: `mkdir`
  - For example: `mkdir /tmp/MUSIC` will create a directory 'Music' under '/tmp' directory
- You can also create more than one directory at a time.

Removing Directories

14

- To remove a directory, use the command: `rmdir directoryname`
  - Example: `rmdir mydirectory` will delete the directory mydirectory
- **Tip**: Ensure that there is no file / sub-directory under the directory that you want to delete. Delete the files/sub-directory first before deleting the parent directory.

Renaming Directory

15

- The 'mv' (move) command (covered earlier) can also be used for renaming directories.
  - Use the given format: `mv directoryname newdirectoryname`

Other Important Commands

16

- **The Man command:**
  - Man stands for manual which is a reference book of a Linux operating system. It is similar to HELP file found in popular software.
  - To get help on any command that you do not understand, you can type: `man`
  - The terminal would open the manual page for that command.
    - For an example, if we type *man man* and hit enter; terminal would give us information on man command
- **The History command:**
  - History command shows all the commands that you have used in the past for the current terminal session.
  - This can help you refer to the old commands you have entered and re-used them in your operations again.
- **The Clear command:**
  - This command clears all the clutter on the terminal and gives you a clean window to work on, just like when you launch the terminal.

Pasting commands into the terminal

17

- Many times you would have to type in long commands on the Terminal.
- Well, it can be annoying at times, and if you want to avoid such a situation then copy, pasting the commands can come to rescue.
- For copying, the text from a source, you would use **Ctrl + c**, but for pasting it on the Terminal, you need to use **Ctrl + Shift + V**.
- You can also try **Shift + Insert or select Edit>Paste on the menu**
- NOTE: With Linux upgrades, these shortcuts keep changing. You can set your preferred shortcuts via Terminal> Edit> Keyboard Shortcuts.

Printing in Linux

18

- Let's try out some easy commands which **can print files** in a format you want. What more, your original file does not get affected at all by the formatting that you do. Let us learn about these commands and their use
- **'pr' command**
  - This command helps in formatting the file for printing on the terminal. There are many options available with this command which help in making desired format changes on file.

‘pr’ commands

19

- The most used **'pr'** options are listed below:

Option	Function
-x	Divides the data into 'x' columns
-h "header"	Assigns "header" value as the report header
-t	Does not print the header and top/bottom margins
-d	Double spaces the output file
-n	Denotes all line with numbers
-l page length	Defines the lines (page length) in a page. Default is 56
-o margin	Formats the page by the margin number

Formatting content with ‘pr’ commands

20

- **Dividing data into columns:**
  - To arrange contents of a file into columns, the syntax would be: `pr -x filename`
    - The '-x' option with the 'pr' command divides the data into x columns.
- **Assigning a header:**
  - To assign a header to the contents of a file: `pr -h "Header" Filename`
    - The '-h' options assigns "header" value as the report header
- **Denoting all lines with numbers**
  - `The syntax to denote lines with numbers: pr -n Filename`
    - This command denotes all the lines in the file with numbers.
- These are some of the 'pr' command options that you can use to modify the file format.

Printing a file

21

- Once you are **done with the formatting**, and it is time for you to get a **hard copy** of the file, you need to use the following command:

  `lp Filename` or `lpr Filename`
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- In windows, the installation of a program is typically done by running the **setup.exe** file.
  - The "setup.exe" installation bundle contains the program as well various dependent components required to run the program correctly.
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- Any dependent components will have to be installed separately which are usually available as packages themselves.
- You can use the **apt** commands to install or remove a package. Let's update all the installed packages in our system using command: `sudo apt-get update`
- The easy and popular way to install programs on Ubuntu is by using the Software center
  - Most of the software packages are available on it and it is far more secure than the files downloaded from the internet.

Sending E-mails

23

- For sending mails through a terminal, you will need to install packages 'mailutils'.
- The command syntax is: `sudo apt-get install packagename`
- Once done, you can then use the following syntax for sending an email: `mail -s 'subject' -c 'cc-address' -b 'bcc-address' 'to-address‘`
- Press Cntrl+D you are finished writing the mail.
  - The mail will be sent to the mentioned address.

Summary

24

- You can format and print a file directly from the terminal. The formatting you do on the files does not affect the file contents.
- In Linux, software is installed in the form of packages. A package contains the program itself. Any dependent component needs to be downloaded separately.
- You can also send e-mails from terminal using the **'mail' command.**

DSM2274N/2294N – Lecture 4a

Basic Linux commands

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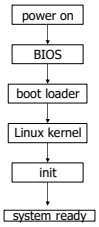
DSM2274N/DSM2294N – Lecture 4b

Linux – Boot process

Linux Start-Up Flow

- BIOS
  - Checks memory, loads options from non-volatile memory, checks for boot devices, loads MBR of boot device and executes it
- MBR
  - Contains a "boot loader" and the partition table
  - Traditionally set up by LILO/GRUB
- Boot loader
  - Loads the compressed kernel image into memory
  - The kernel uncompress itself and starts…
- Init process
  - Configuration file /etc/inittab
  - run levels



System Startup

How does a computer startup?

- Booting is a bootstrapping process that starts operating systems when the user turns on a computer system
- A boot sequence is the set of operations the computer performs when it is switched on that load an operating system

Booting sequence

1. Turn on
2. CPU jump to address of BIOS (0xFFFF0)
3. BIOS runs POST (Power-On Self Test)
4. Find bootable devices
5. Loads and execute boot sector from MBR
6. Load OS

BIOS (Basic Input / Output System)

- BIOS refers to the software code run by a computer when first powered on
- The primary function of BIOS is code program embedded on a chip that recognises and controls various devices that make up the computer.

BIOS on board

BIOS on screen

Boot loader

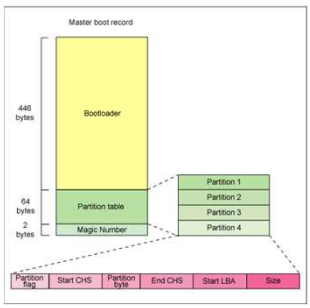
7

MBR (Master Boot Record)

- OS is booted from a hard disk, where the Master Boot Record (MBR) contains the primary boot loader
- The MBR is a 512-byte sector, located in the first sector on the disk (sector 1 of cylinder 0, head 0)
- After the MBR is loaded into RAM, the BIOS yields control to it.
- The first 446 bytes are the primary boot loader, which contains both executable code and error message text
- The next sixty-four bytes are the partition table, which contains a record for each of four partitions
- The MBR ends with two bytes that are defined as the magic number (0xAA55). The magic number serves as a validation check of the MBR

8

MBR (Master Boot Record)



9

Boot loader

- Boot loader could be more aptly called the kernel loader. The task at this stage is to load the Linux kernel
- Optional, initial RAM disk
- GRUB and LILO are the most popular Linux boot loader.
- Other boot loaders (for various OS):
  - Bootman, NTLDR, XOSL, BootX, loadlin, Gujin, Boot Camp, Syslinux, GAG

10

GRUB: GRand Unified Bootloader

- GRUB is an operating system independent boot loader
- A multiboot software packet from GNU
- Flexible command line interface
- File system access
- Supports multiple executable format
- Supports diskless system
- Download OS from network

11

GRUB boot process

1. The BIOS finds a bootable device (hard disk) and transfers control to the master boot record
2. The MBR contains GRUB stage 1. Given the small size of the MBR, Stage 1 just load the next stage of GRUB
3. GRUB Stage 1.5 is located in the first 30 kilobytes of hard disk immediately following the MBR. Stage 1.5 loads Stage 2.
4. GRUB Stage 2 receives control, and displays to the user the GRUB boot menu (where the user can manually specify the boot parameters).
5. GRUB loads the user-selected (or default) kernel into memory and passes control on to the kernel.

12

LILO: LInux LOader

- Not depend on a specific file system
- Can boot from harddisk and floppy
- Up to 16 different images
- Must change LILO when kernel image file or config file is changed

Kernel

Kernel image

- The kernel is the central part in most computer operating systems because of its task, which is the management of the system's resources and the communication between hardware and software components
- The initial program was an exact image of the kernel, stored on disc, of what needed to be loaded into memory and "called" directly to get the kernel going.
- The kernel is always stored on memory until computer is turned off
- A kernel image is not an executable kernel, but a compressed version of the kernel.
  - zImage size less than 512 KB
  - bzImage size greater than 512 KB

Tasks of kernel

- Process management
- Memory management
- Device management
- System call

Init process

- The first thing the kernel does is to execute init program
- Init is the root/parent of all processes executing on Linux
- The first processes that init starts is a script /etc/rc.d/rc.sysinit
- Based on the appropriate run-level, scripts are executed to start various processes to run the system and make it functional

The Linux Init Processes

- The init process is identified by process id "1"
- Init is responsible for starting system processes as defined in the /etc/inittab file
- Init typically will start multiple instances of "getty" which waits for console logins which spawn one's user shell process
- Upon shutdown, init controls the sequence and processes for shutdown

Inittab file

- The inittab file describes which processes are started at bootup and during normal operation
  - /etc/init.d/boot
  - /etc/init.d/rc
- The computer will be booted to the runlevel as defined by the initdefault directive in the /etc/inittab file
  - id:5:initdefault:

Runlevels

- A runlevel is a software configuration of the system which allows only a selected group of processes to exist
- The processes spawned by init for each of these runlevels are defined in the /etc/inittab file
- Init can be in one of eight runlevels: 0-6, S or M

Runlevels

Runlevel	Scripts Directory (Red Hat/Fedora Core)	State
0	/etc/rc.d/rc0.d/	shutdown/halt system
1	/etc/rc.d/rc1.d/	Single user mode
2	/etc/rc.d/rc2.d/	Multiuser with no network services exported
3	/etc/rc.d/rc3.d/	Default text/console only start. Full multiuser
4	/etc/rc.d/rc4.d/	Reserved for local use. Also X-windows (Slackware/BSD)
5	/etc/rc.d/rc5.d/	XDM X-windows GUI mode (Redhat/System V)
6	/etc/rc.d/rc6.d/	Reboot
s or S		Single user/Maintenance mode (Slackware)
M		Multiuser mode (Slackware)

DSM2274N/2294N – Lecture 4c

Linux – File permissions

File Permissions in Linux

- Linux is a clone of UNIX, the **multi-user operating system** which can be accessed by many users simultaneously.
- Linux can also be used in mainframes and servers without any modifications.
- But this raises security concerns as an unsolicited or **malign user** can **corrupt, change or remove crucial data**.
- For effective security, Linux divides authorization into 2 levels.
  - Ownership
  - Permission

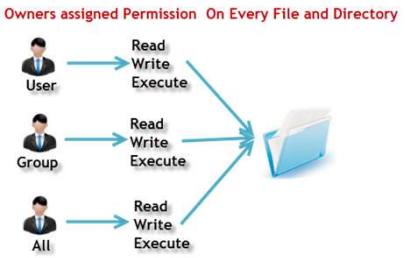
Ownership of Linux files

- Every file and directory on your Unix/Linux system is assigned 3 types of owners:
  - **User**
    - A user is the owner of the file. By default, the person who created a file becomes its owner. Hence, a user is also sometimes called an owner.
  - **Group**
    - A user-group can contain multiple users. All users belonging to a group will have the same access permissions to the file.
    - Suppose you have a project where a number of people require access to a file. Instead of manually assigning permissions to each user, you could add all users to a group, and assign group permission to file such that only this group members and no one else can read or modify the files.
  - **Other**
    - Any other user who has access to a file. This person has neither created the file, nor he belongs to a usergroup who could own the file. Practically, it means everybody else. Hence, when you set the permission for others, it is also referred as set permissions for the world.
- How does **Linux distinguish** between these three user types so that a user 'A' cannot affect a file which contains some other user 'B's' vital information/data?
  - This is where **Permissions** set in, and they define **user behavior**.

Permissions

- Every file and directory in your UNIX/Linux system has following 3 permissions defined for all the 3 owners:
  - **Read**
    - This permission give you the authority to open and read a file. Read permission on a directory gives you the ability to lists its content.
  - **Write**
    - The write permission gives you the authority to modify the contents of a file. The write permission on a directory gives you the authority to add, remove and rename files stored in the directory.
    - Consider a scenario where you have to write permission on file but do not have write permission on the directory where the file is stored. You will be able to modify the file contents. But you will not be able to rename, move or remove the file from the directory.
  - **Execute**
    - In Windows, an executable program usually has an extension ".exe" and which you can easily run. In Unix/Linux, you cannot run a program unless the execute permission is set.
    - If the execute permission is not set, you might still be able to see/modify the program code(provided read & write permissions are set), but not run it.

Permissions applied



Permissions in action

- Type `ls -l` on terminal:
  - Result: `-rw-rw-r-- 1 home home`
- Here, we have highlighted **'-rw-rw-r--'** and this weird looking code is the one that tells us about the permissions given to the owner, user group and the world.
  - The first '-' implies that we have selected a file (if it were a directory, **d** would have been shown).
- The subsequent characters are pretty easy to remember.
  - **r** = read permission
    w = write permission
    x = execute permission
    - = no permission

Permissions explained

- Result: `-rw-rw-r-- 1 home home`
  - The first part of the code is **'rw-'**. This suggests that the owner 'Home' can:
    - Read the file
    - Write or edit the file
    - He cannot execute the file since the execute bit is set to '-'.
    - By design, many Linux distributions like Fedora, CentOS, Ubuntu, etc. will add users to a group of the same group name as the user name. Thus, a user 'tom' is added to a group named 'tom'
  - The second part is **'rw-'**. It for the user group 'Home' and group-members can:
    - Read the file
    - Write or edit the file
  - The third part is for the world which means any user. It says **'r--'**. This means the user can only:
    - Read the file

Changing file / directory permissions

- Say you do not want your colleague to see your personal images.
- This can be achieved by changing file permissions.
- We can use the **'chmod'** command which stands for 'change mode'.
- Using the command, we can set permissions (read, write, execute) on a file/directory for the owner, group and the world.
- **Syntax: `chmod permissions filename`**
- There are 2 ways to use the command:
  - **Absolute mode**
  - **Symbolic mode**

Absolute (Numeric) Mode

- In this mode, file permissions are **not represented as characters but a three-digit OCTAL number**.
- The table below gives numbers for all for permissions types.

Number	Permission Type	Symbol
0	No Permission	---
1	Execute	--x
2	Write	-w-
3	Execute + Write	-wx
4	Read	r--
5	Read + Execute	r-x
6	Read +Write	rw-
7	Read + Write +Execute	rwx

Absolute (Numeric) Mode in action

- Let's see the chmod command in action:
  `ls -l sample`
  `-rw-rw-r-` (this checks the current file permission)
  `chmod 764 sample`
  `ls -l sample`
- In the above-given terminal window, we have changed the permissions of the file 'sample to '764'.
  - '764' absolute code says the following:
    - Owner can read, write and execute
    - Usergroup can read and write
    - World can only read
- This is shown as `rwxrw--r-`
  - This is how you can change the permissions on file by assigning an absolute number.

Symbolic Mode

- In the Absolute mode, you change permissions for all 3 owners.
- In the symbolic mode, you can modify permissions of a specific owner.
- It makes use of mathematical symbols to modify the file permissions:

Operator	Description
+	Adds a permission to a file or directory
-	Removes the permission
=	Sets the permission and overrides the permissions set earlier.

- The various owners are represented as:

User Denotations	
u	user/owner
g	group
o	other
a	all

- We will not be using permissions in numbers like 755 but characters like rwx.

Changing Ownership and Group

- For changing the ownership of a file/directory, you can use the following command: **`chown user`**
- In case you want to change the user as well as group for a file or directory use the command: **`chown user:group filename`**
- Let's see this in action:
  - Check the current file ownership: `ls -l`
  - Change the file name to n100: **`sudo chown n100 sample.txt`**
  - Change user and group: **`sudo chown root:root sample.txt`**
- In case you want to change group-owner only, use the command: **`chgrp group_name filename`**
  - **'chgrp'** stands for change group.

Tips for Changing Ownership and Group

- The file /etc/group contains all the groups defined in the system
- You can use the command '**groups**' to find all the groups you are a member of
- You can use the command '**newgrp**' to work as a member a group other than your default group
- You cannot have 2 groups owning the same file.
- You do not have nested groups in Linux. One group cannot be sub-group of other
- x- eXecuting a directory means Being allowed to "enter" a dir and gain possible access to sub-dirs
- There are other permissions that you can set on Files and Directories which will be covered in a later advanced tutorial

Summary

- Linux being a multi-user system uses permissions and ownership for security.
- There are three user types on a Linux system viz. User, Group and Other
- Linux divides the file permissions into read, write and execute denoted by r,w, and x
- The permissions on a file can be changed by 'chmod' command which can be further divided into Absolute and Symbolic mode
- The 'chown' command can change the ownership of a file/directory. Use the following commands: chown user file or chown user:group file
- The 'chgrp' command can change the group ownership chrgrp group filename
- What does x - eXecuting a directory mean? A: Being allowed to "enter" a dir and gain possible access to sub-dirs.